UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Friday 6 June 2025 09.00 to 12.00 (*3 hours plus any pre-agreed individual adjustment*)

Paper PHA/1

PHARMACOLOGY EXTENDED ANSWER (PAPER 1)

*All candidates must provide **one answer** for **each question** from the choice of a or b. You therefore need to provide **six** answers in total*

Diagrams or other handwritten material:
Where appropriate (e.g., for formulae, equations or diagrams) you may hand write your answers using the Inspera Scan Paper provided. You should not copy and paste images from any other documents.

Please Note:
- *Inspera Scan question codes MUST be noted during the exam.*
- *Please ensure that each Inspera Scan page is labelled appropriately, including your Blind Grade Number.*
- *Inspera Scan pages MUST be placed in the Inspera Scan folder at the completion of the exam and left on your desk.*
- *Inspera scan pages will be scanned by administrators after the completion of the exam.*

Stationery requirements:
- *Rough work pad*
- *Inspera Scan pages*
- *Inspera Scan Folder*

1. **Either**

   a. Explain how different anticholinesterase agents act and are used therapeutically.

   or

   b. Describe how drug leads can be optimised to improve their potency and efficacy.

2. **Either**

   a. Discuss the mechanisms of membrane-targeting antimicrobial drugs.

   or

   b. Describe bacterial antibiotic resistance mechanisms.

3. **Either**

   a. Explain the different therapeutic strategies to target growth factor signalling pathways in cancer.

   or

   b. Discuss the role of the immune system in cancer development and treatment.

4. **Either**

   a. Describe the anti-inflammatory actions of corticosteroids.

   or

   b. Using specific examples, discuss how immunosuppressive drugs enable organ transplantations.

5. **Either**

   a. Explain how drugs may affect the force of contraction of the heart.

   or

   b. Discuss the mechanisms of action of local anaesthetics.

6. **Either**

   a. Discuss hormones of the female reproductive system and describe how and why they are pharmacologically targeted.

   or

   b. Describe drug targets underlying the pharmacological treatment of anxiety.

END OF PAPER